

AWS Capstone Project Submission

Student Name: Sumaiya Sara

Project: AWS Cloud Architecting Capstone Project

Status: Completed Successfully

Project Overview

This project was completed using AWS cloud services to build a secure, highly available web application architecture. The application was deployed using EC2 instances in private subnets, an Application Load Balancer, Auto Scaling Group, RDS MySQL database, and AWS Secrets Manager.

AWS Services Used

- Amazon EC2
- Amazon RDS MySQL
- Application Load Balancer (ALB)
- Auto Scaling Group (ASG)
- AWS Secrets Manager
- Amazon VPC
- Security Groups
- Private and Public Subnets

Completed Tasks

- Created MySQL RDS database in private subnets
- Configured AWS Secrets Manager for secure credentials storage
- Created Application Load Balancer in public subnets
- Configured Auto Scaling Group across two Availability Zones
- Launched EC2 instances using Project Launch Template
- Configured security groups between ALB, EC2, and RDS
- Imported SQL country dataset into RDS database
- Successfully tested application connectivity
- Verified application accessibility through ALB DNS
- Received full marks on project submission

Architecture Flow

Internet → Application Load Balancer → EC2 Instances (Private Subnets) → RDS MySQL Database → Query Results Returned to User

Submission Notes

The project was fully completed and successfully submitted through the AWS Academy lab environment. The grading report confirmed that the application was reachable, the

2. Load Balancer Created

Account ID: 9092-3291-7169

voclabs/user4775005=sumaiya.sa...

United States (N. Virginia)

Search

Option+S

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 3 - optional

Integrate with other services

Step 4 - optional

Configure group size and scaling

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Load balancing Info

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

☐ No load balancer

Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ Attach to an existing load balancer

Choose from your existing load balancers.

☐ Attach to a new load balancer

Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers to attach

☒ Choose from your load balancer target groups

This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ Choose from Classic Load Balancers

Existing load balancer target groups

Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

capstone-tg | HTTP

Application Load Balancer: capstone-alb

VPC Lattice integration options Info

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

Select VPC Lattice service to attach

☒ No VPC Lattice service

VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

☐ Attach to VPC Lattice service

Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

Create new VPC Lattice service

Application Recovery Controller (ARC) zonal shift - new Info

During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

☐ Enable zonal shift

New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

[Always enabled](#)

Additional health check types - optional Info

☐ Turn on Elastic Load Balancing health checks **Recommended**

Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☐ Turn on VPC Lattice health checks

VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ Turn on Amazon EBS health checks

EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period Info

This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

300 seconds

Cancel

Skip to review

Previous

Next

Account ID: 9092-3291-7169
voclabs/user4775005=sumaiya.sa...

United States (N. Virginia)

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
Choose launch template or configuration

Step 2
Choose instance launch options

Step 3 - optional
Create Auto Scaling group

Step 4 - optional
Configure group size and scaling

Step 5 - optional
Add notifications

Step 6 - optional
Add tags

Step 7
Review

Configure group size and scaling - optional [Info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

2

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

4

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies

☒ Target tracking scaling policy

☐ Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

Target Tracking Policy

Metric type [Info](#)

Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization

Target value

50

Instance warmup [Info](#)

300 seconds

☐ Disable scale in to create only a scale-out policy

Instance maintenance policy [Info](#)

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

Mixed behavior

☒ No policy

☐ Launch before terminating

Prioritize availability

☐ Launch new instances and wait for them to be

Control costs

☐ Terminate and launch instances at the same time. This allows you to go below your desired

Flexible

☐ Custom behavior

☒ For rebalancing events, new instances will launch before terminating others. For

☐ Launch new instances and wait for them to be

☐ Terminate and launch instances at the same time. This allows you to go below your desired

☐ Set custom values for the minimum and maximum amount of available capacity. This

Create Auto Scaling group

Instance maintenance policy [Info](#)

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

Mixed behavior

- ☒ **No policy**
For rebalancing events, new instances will launch before terminating others. For all other events, instances terminate and launch at the same time.

Prioritize availability

- ☐ **Launch before terminating**
Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.

Control costs

- ☐ **Terminate and launch**
Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.

Flexible

- ☐ **Custom behavior**
Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.

Additional capacity settings

Capacity Reservation preference [Info](#)

Select whether you want Auto Scaling to launch instances into an existing Capacity Reservation or Capacity Reservation resource group.

- ☒ **Default**
Auto Scaling uses the Capacity Reservation preference from your launch template.
- ☐ **None**
Instances will not be launched into a Capacity Reservation.
- ☐ **Capacity Reservations only**
Instances will only be launched into a Capacity Reservation. If capacity isn't available, the instances fail to launch.
- ☐ **Capacity Reservations first**
Instances will attempt to launch into a Capacity Reservation first. If capacity isn't available, instances will run in On-Demand capacity.

Additional settings

Instance scale-in protection

If protect from scale in is enabled, newly launched instances will be protected from scale in by default.

- ☐ **Enable instance scale-in protection**

Monitoring [Info](#)

- ☐ **Enable group metrics collection within CloudWatch**

Default instance warmup [Info](#)

The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

- ☐ **Enable default instance warmup**

Auto Scaling group deletion protection - new [Info](#)

Configure how this Auto Scaling group can be deleted.

- ☒ **None (default)**
The Auto Scaling group can be deleted.
- ☐ **Prevent force deletion**
The Auto Scaling group cannot be force deleted. To delete the group, all instances must be detached or terminated, and all warm pools must be deleted.
- ☐ **Prevent all deletion**
The Auto Scaling group cannot be deleted.

Placement group [Info](#)

Choose a placement group to determine how instances are placed on underlying hardware.

None

[Create a placement group](#)

Cancel

Skip to review

Previous

Next

3. Auto Scaling Group Running

aws [Search] [Option+S] United States (N. Virginia) Account ID: 9092-3291-7169 voclabs/user4775005=sumaiya.sa...

EC2 > Auto Scaling groups

Auto Scaling groups (1) Info Last updated 2 minutes ago Launch configurations Launch templates Actions Create Auto Scaling group

Search your Auto Scaling groups < 1 >

☐	Name	Status - new	Launch template/config...	Insta...	Instance health...	Desired ca...	Min	M
☐	capstone-as	Updating capacity..	Project-LT	Version Default	0	-	2	1 4

4. EC2 Instances Running

Instances (2) Info Connect Instance state Actions Launch instances

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive) < 1 >

☐	Name	Instance ID	Instance s...	Instance ...	Status check	Availability...	Public IPv4 DNS
☐	capstone-...	i-01c296846298c47cd	Running	t2.micro	2/2 checks pa	us-east-1a	-
☐	capstone-...	i-07b217cd88a9b8e45	Running	t2.micro	2/2 checks pa	us-east-1b	-

5. Working Website Homepage

Example Social Research Organization

[About Us](#) [Contact Us](#) [Query](#)

Welcome to our data query site. You can get data from countries all over the world to use in your research.

We provide data for a variety of areas including basic demographics and development statistics.

About Us



Shirley Rodriguez

Our site got started when Shirley Rodriguez found that she was frequently looking up data from a variety of databases.

Shirley decided to start sharing some of this data with other social researchers.

Contact Us

6. Database Query Results

[Pick another query](#)

Country Name	Population	Urban Population
Afghanistan	26697430	5771984
Albania	3071856	1280964
Algeria	30533827	18259229
American Samoa	57625	51171
Andorra	64634	59722
Angola	13926373	6823923
Antigua and Barbuda	77656	24928
Argentina	36930709	33274569
Armenia	3076098	2002540
Aruba	90271	42157
Australia	19153000	16701416
Austria	8011566	5271610
Azerbaijan	8048600	4120883
Bahamas, The	297651	244074
Bahrain	638193	564163
Bangladesh	129592275	30583777
Barbados	267511	97106
Belarus	10005000	6993495
Belgium	10251250	9953964
Belize	249800	119404
Benin	6517810	2496321
Bermuda	62100	62100
Bhutan	571262	145101
Bolivia	8307248	5133879
Bosnia and Herzegovina	3693698	1595678
Botswana	1757925	935216
Brazil	174425387	141633414
Brunei Darussalam	327036	232523
Bulgaria	8170172	5629249
Burkina Faso	12294012	2040806
Burundi	6374347	529071
Cambodia	12446949	2103534
Cameroon	15678269	7823456
Canada	30769700	24461912
Cape Verde	437238	233485

Capstone Project

Due No Due Date Points 35 Submitting an external tool

AWS

Used \$0 of \$60

01:11

▶ Start Lab

■ End Lab

ⓘ AWS Details

ⓘ Details

↺ Reset

✕

Submit

Submission Report

Grades

submissions recorded for this lab.

Ending your session

Reminder: This is a long-lived lab environment. Data is retained until either you use the allocated budget or the course end date is reached (whichever occurs first).

To preserve your budget when you are finished for the day or when you are finished actively working on the assignment for the time being, do the following:

16. At the top of this page, choose **End Lab**, and then choose Yes to confirm that you want to end the lab.

A message appears: *Ended AWS Lab Successfully*

Note: Choosing **End lab** in this lab environment will not delete the resources that you have created. They will still be there the next time you choose **Start Lab** (for example, on another day).

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Total score

35/35

[Task 1] Load Balancer is created

5/5

[Task 2] Application is reachable

5/5

[Task 3] Application Server is created in Private Subnets

5/5

[Task 4] RDS MySQL Database is created

5/5

[Task 5] Database uses Secrets Manager

5/5

[Task 6] Database is not public

5/5

[Task 7] EC2 Instances are launched in Auto Scaling group

5/5